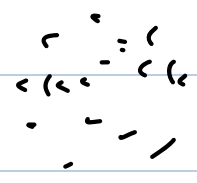
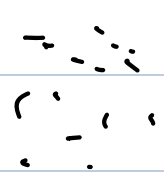


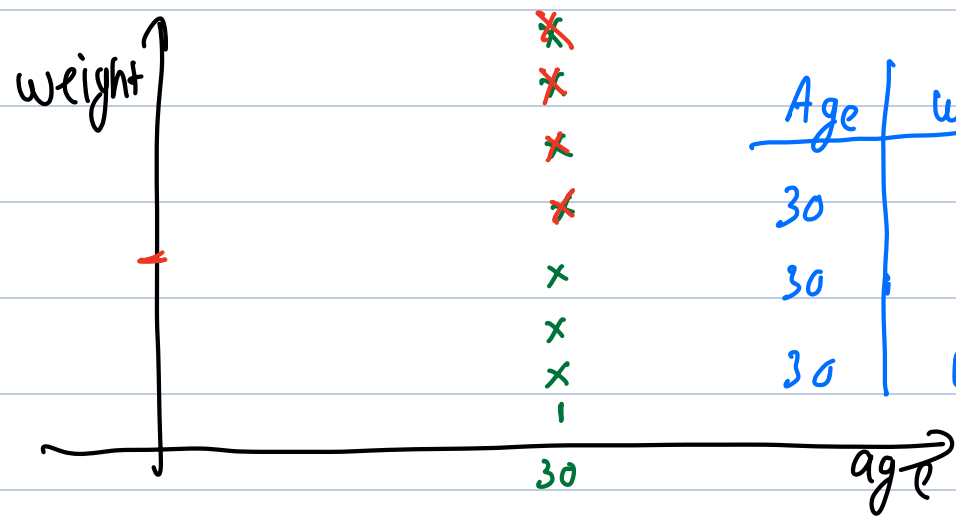
9:31

PCA



→ 2D

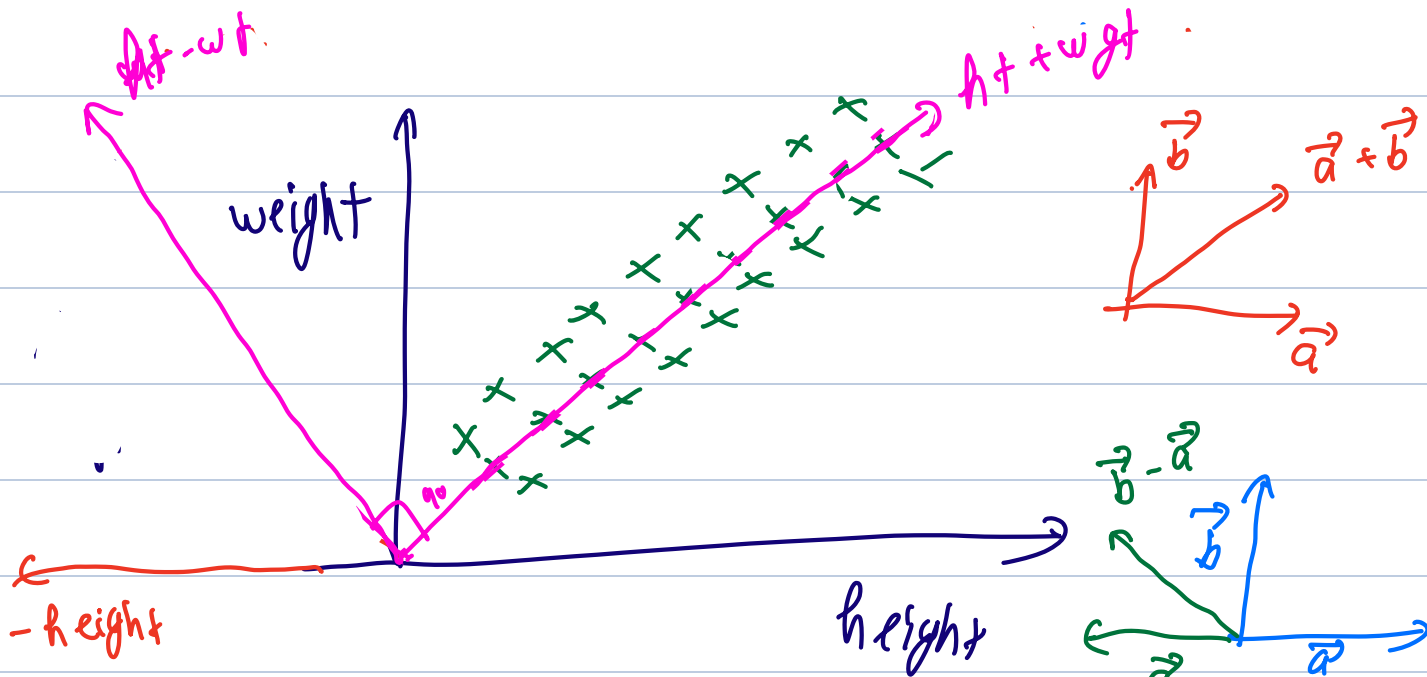
→ 3D



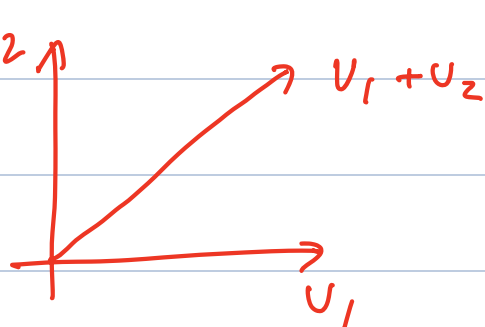
Age	Weight	Diab ^d
30	5	No
30	8	yes
30	10	yes



Age	Weight
30	5
29.9	8
30	10
30.1	12
29.9	14

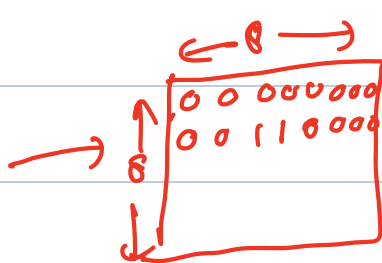
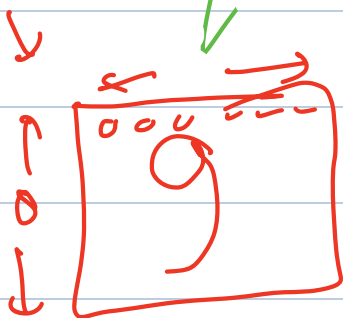


height	weight	$h + wt$	$h - wt$
—	—	—	—
—	—	—	—
—	—	—	—

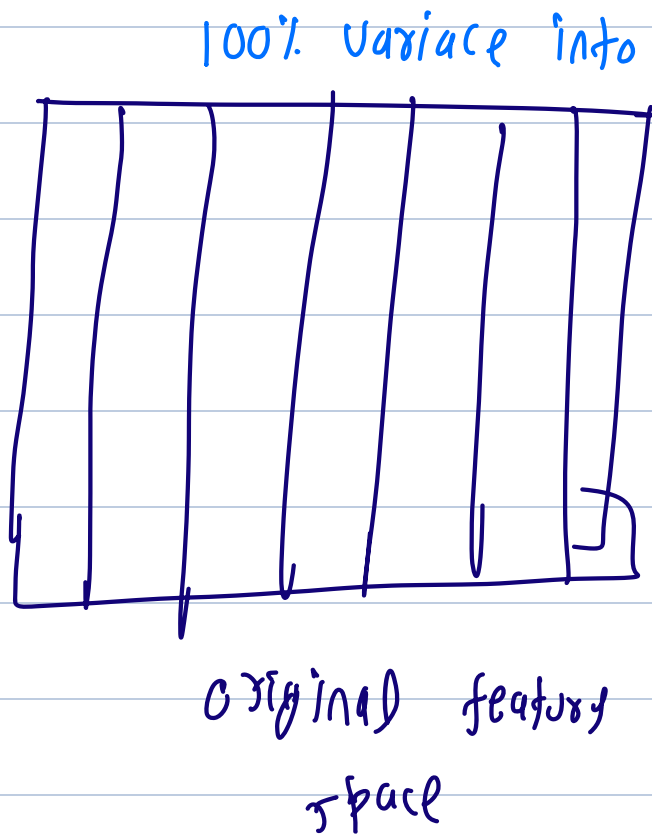
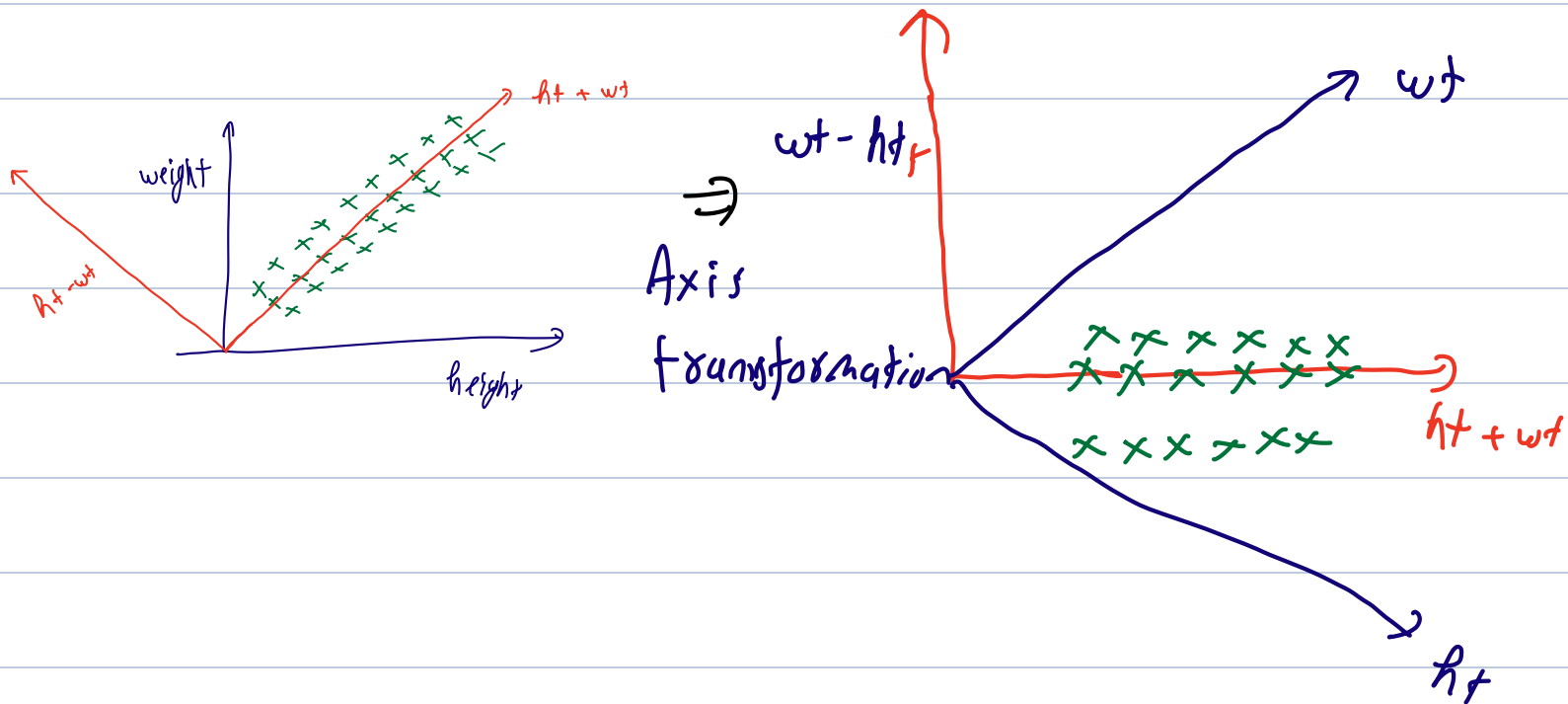


h	w	$h + w$
2	3	5
4	5	9

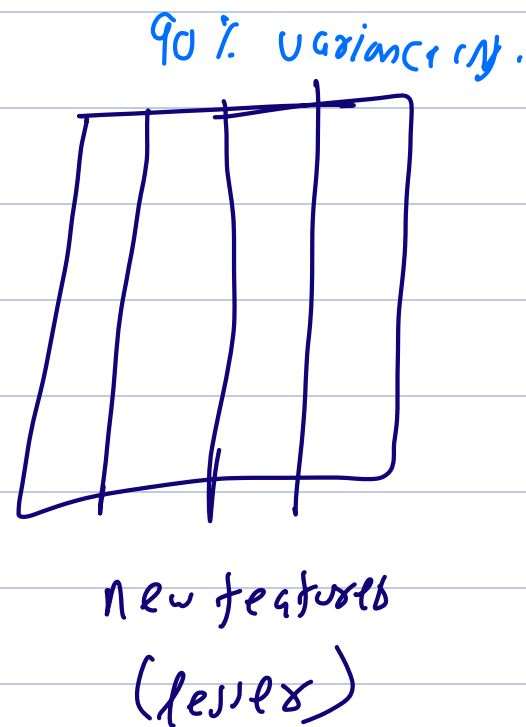
Blackwhite



label	p_0	p_1	p_2	...	p_{63}
9					



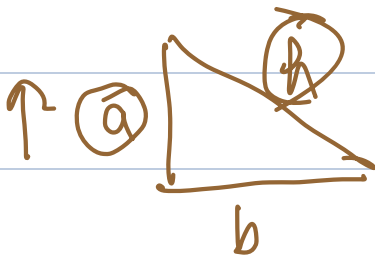
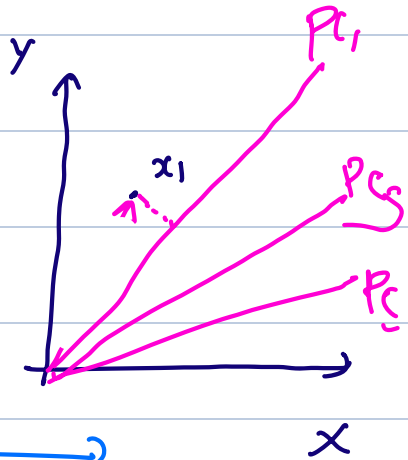
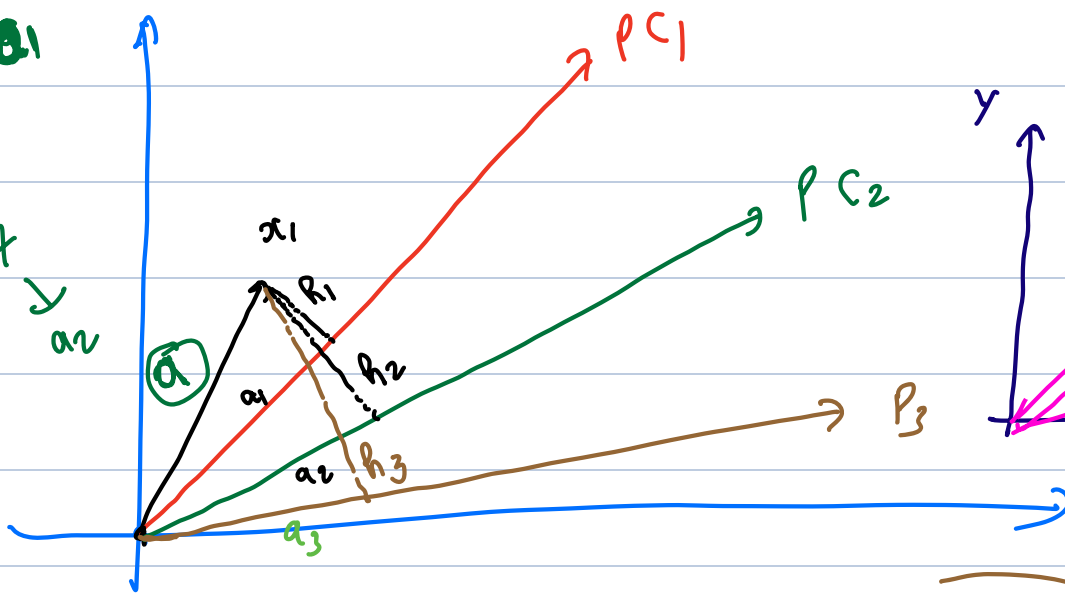
PCA $\rightarrow$



100% variance
 m features $\xrightarrow{\text{PCA}}$ m' (90% variance)
 $m' < m$

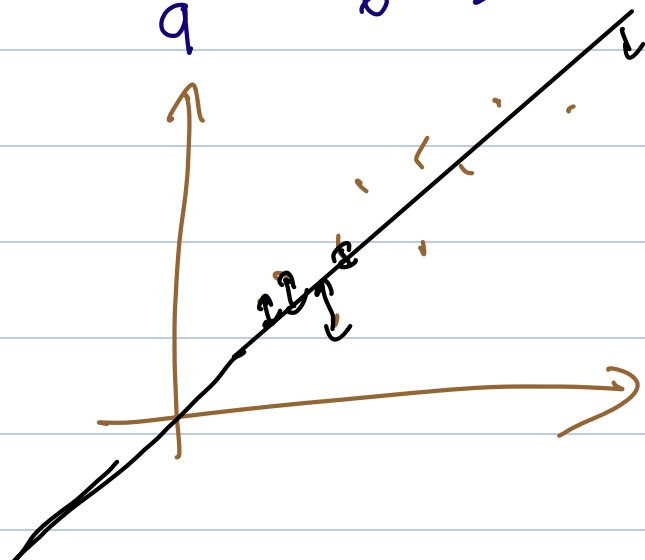
...

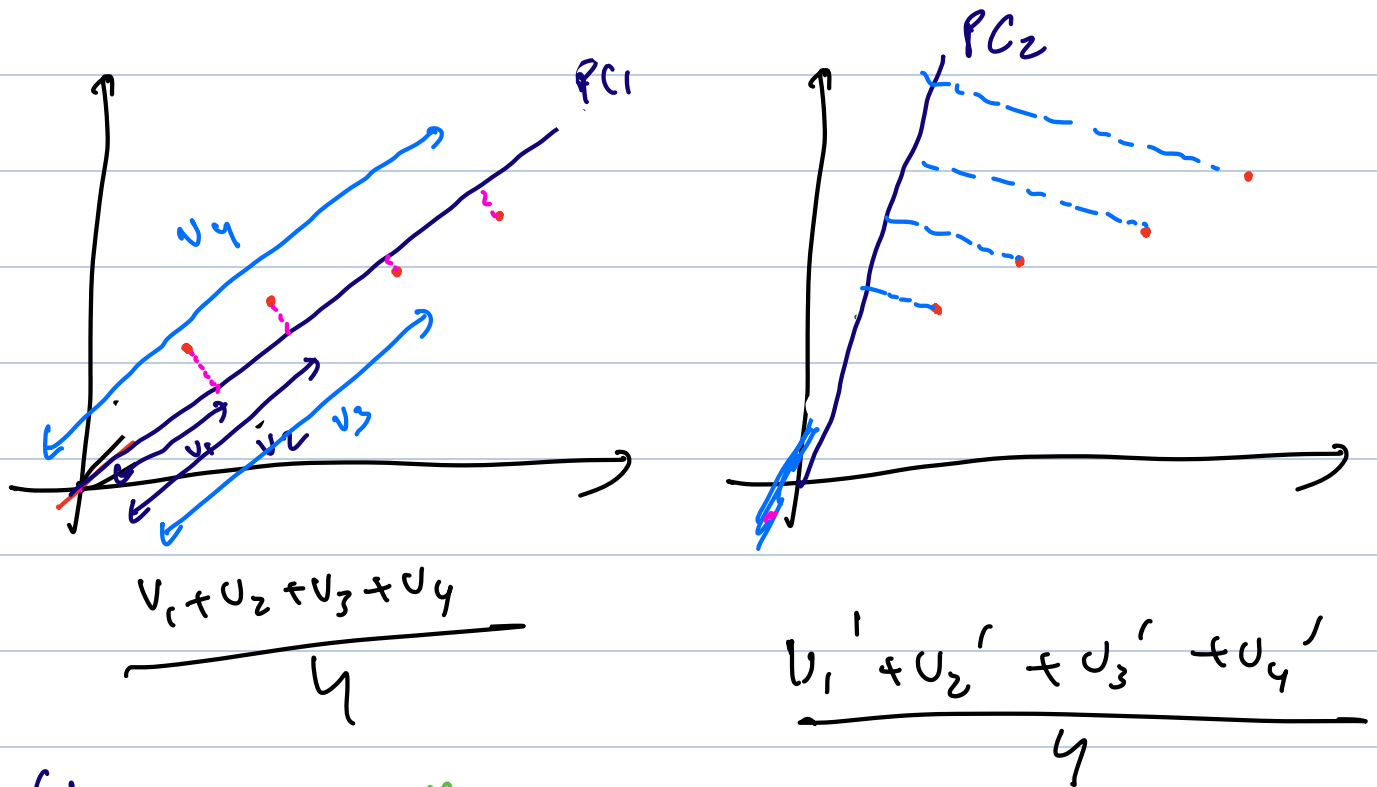
h_1
 $q = 1 + 8 \rightarrow a_1$
 $q = 2 + 7 \rightarrow a_2$



$h^2 = a^2 + b^2$
 $q = 4$ (under a)
 $q = 9$ (under b)
 $z = 5$ (under h)
 $z = 3$ (under b)

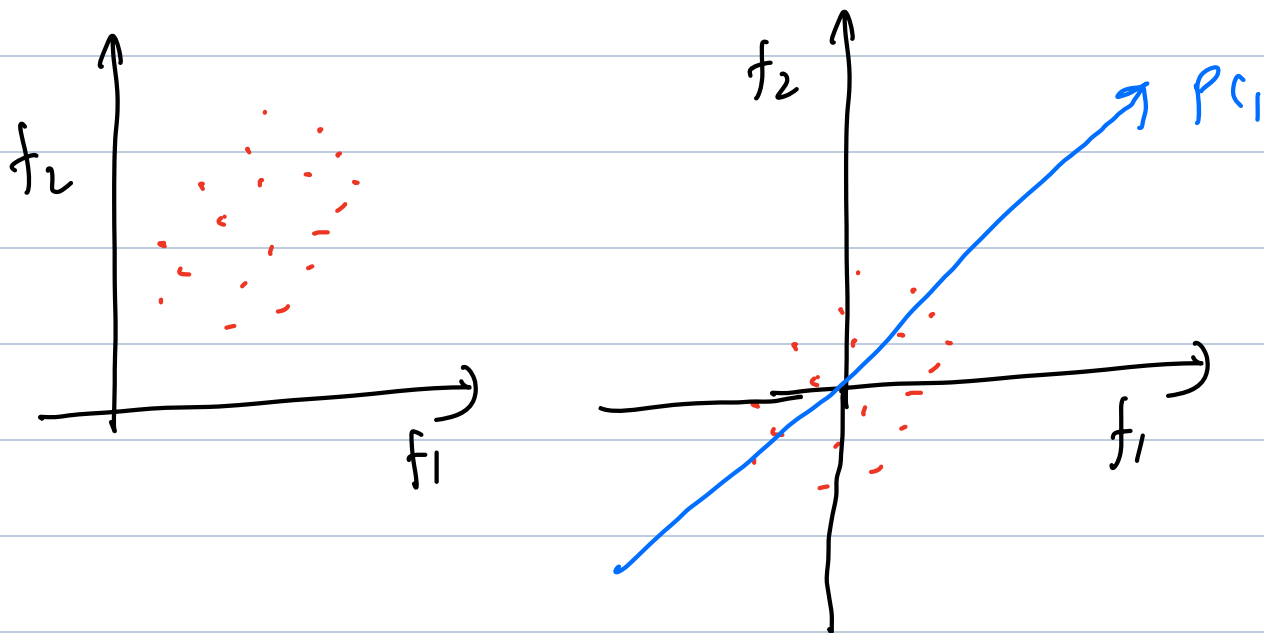
$q = 4 + 5$
 $= 9$





Break. 6:

Mathematical Proof



$$PC_1 : w^T x + w_0 = 0$$

16

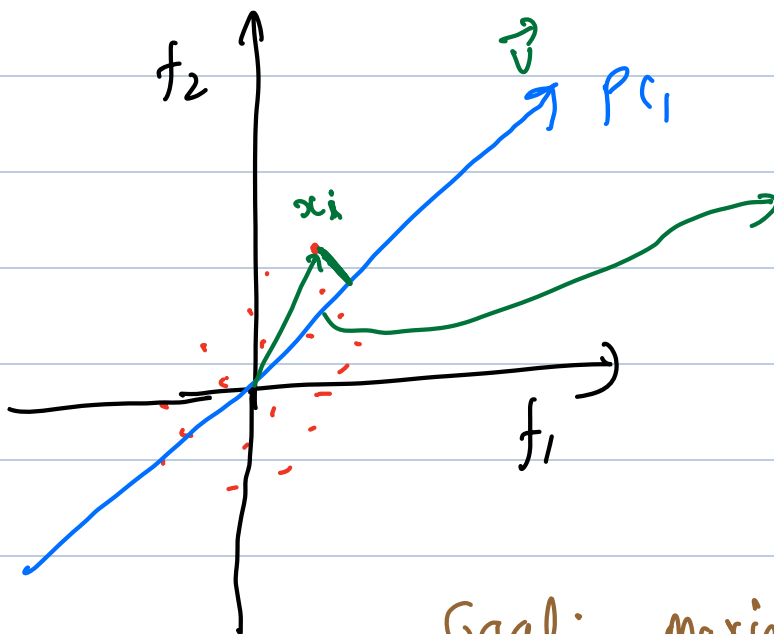
assume $w_0 = 0$

$$w^T x = 0$$

$$v^T x = 0$$

c_1

2



For projection of x_i on $\vec{v}$

$$\Rightarrow \frac{\vec{x}_i \cdot \vec{v}}{\|\vec{v}\|}$$

Goal: maximise length of projection

Calculating all vectors projection on $\vec{v}$

$$\max \frac{1}{n} \sum_{i=1}^n \frac{\vec{x}_i \cdot \vec{v}}{\|\vec{v}\|}$$

$\|\vec{v}\| = 1$
(unit vector)

$$\Rightarrow \max \frac{1}{n} \sum_{i=1}^n \vec{x}_i \cdot \vec{v} \quad \text{s.t.} \quad \|\vec{v}\| = 1$$

constraint optimisation

$$\max_{\vec{v}, \lambda} \frac{1}{n} \sum \vec{x}_i \cdot \vec{v} + \lambda (\|\vec{v}\| - 1)$$

$$\max_{\vec{v}, \lambda} \frac{1}{n} \sum (\vec{x}_i \cdot \vec{v})^2 + \lambda (\|\vec{v}\| - 1)$$

$$\max_{\vec{v}, \lambda} \frac{(x \cdot \vec{v})^2}{n} + \lambda (\|\vec{v}\| - 1)$$

$$A^2 = A^T \cdot A$$

$$\Rightarrow \max_{\vec{v}, \lambda} \frac{(x \cdot v)^T (x \cdot v)}{n} + \lambda (v^T v - 1)$$

$$(AB)^T = B^T \cdot A^T$$

$$\max_{\vec{v}, \lambda} \frac{\vec{v}^T \cdot \frac{X^T \cdot X}{n} \cdot \vec{v}}{\lambda (\vec{v}^T \vec{v} - 1)}$$

	f_1	f_2	f_3
f_1	—	—	—
f_2	—	—	—
f_3	—	—	—

Co-variance

	x_1	x_2	x_3	...	x_n
f_1	—	—	—		—
f_2	—	—	—		—
f_3					
$\vdots$					
f_n					

	f_1	f_2	...	f_n
x_1				
x_2				
x_3				
$\vdots$				
x_n				

$$(x_1 \ x_2 \ \dots \ x_n) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \frac{x_1^2 + x_2^2 + \dots + x_n^2}{n}$$

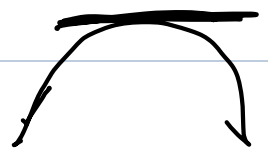
$$\frac{(x_1 - \mu_1)^2 + (x_2 - \mu)^2 + \dots + (x_n - \mu)^2}{n}$$

$$\max_{\vec{v}, \lambda} \frac{\vec{v}^T \cdot \underbrace{X^T \cdot X}_{n} \cdot \vec{v}}{n} + \lambda (\vec{v}^T \vec{v} - 1)$$

$$\max_{\vec{v}, \lambda} \vec{v}^T \cdot \mathbf{V} \cdot \vec{v} + \lambda (\vec{v}^T \vec{v} - 1)$$

$$\mathcal{L} = \vec{v}^T \mathbf{V} \cdot \vec{v} + \lambda (\vec{v}^T \vec{v} - 1)$$

$$\mathcal{L} = \vec{v}^T \mathbf{V} \cdot \vec{v} + \lambda \vec{v}^T \vec{v} - \lambda$$



$$\frac{\partial \mathcal{L}}{\partial \vec{v}} = 2 \mathbf{V} \vec{v} + 2 \lambda \vec{v}$$

$$\frac{\partial L}{\partial \lambda} = U^T U - I$$

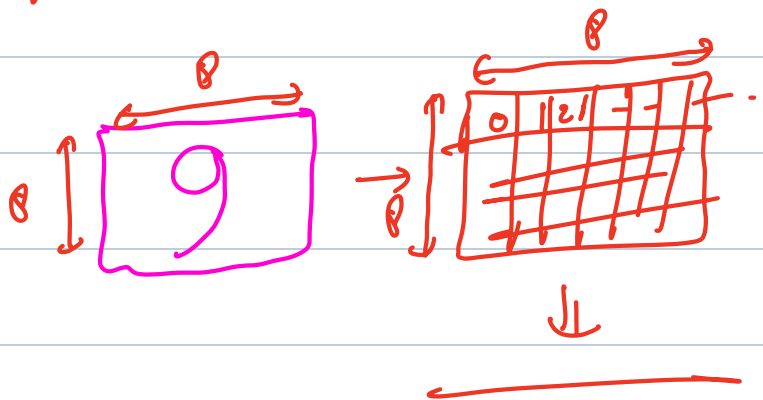
$$U^T U - I = 0$$

$$U^T U = I$$

$$2VU + 2\lambda U = 0$$

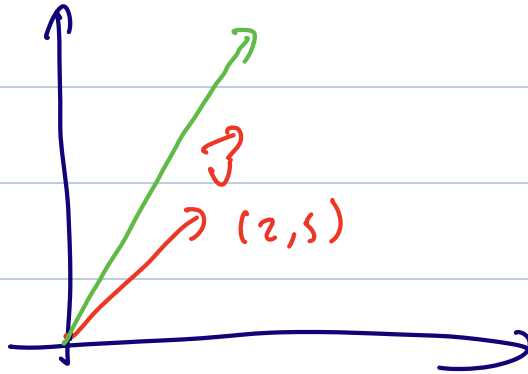
$$\cancel{2}VU = -\cancel{2}\lambda U$$

$$\begin{array}{c}
 VU = -\lambda U \\
 \downarrow \quad \searrow \quad \downarrow \quad \swarrow \\
 \text{matrix} \quad \text{vector} \quad \text{scalar} \quad \text{vector}
 \end{array}$$



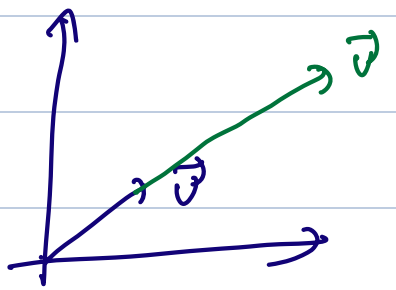
$$V = \begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix}$$

$$u = \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

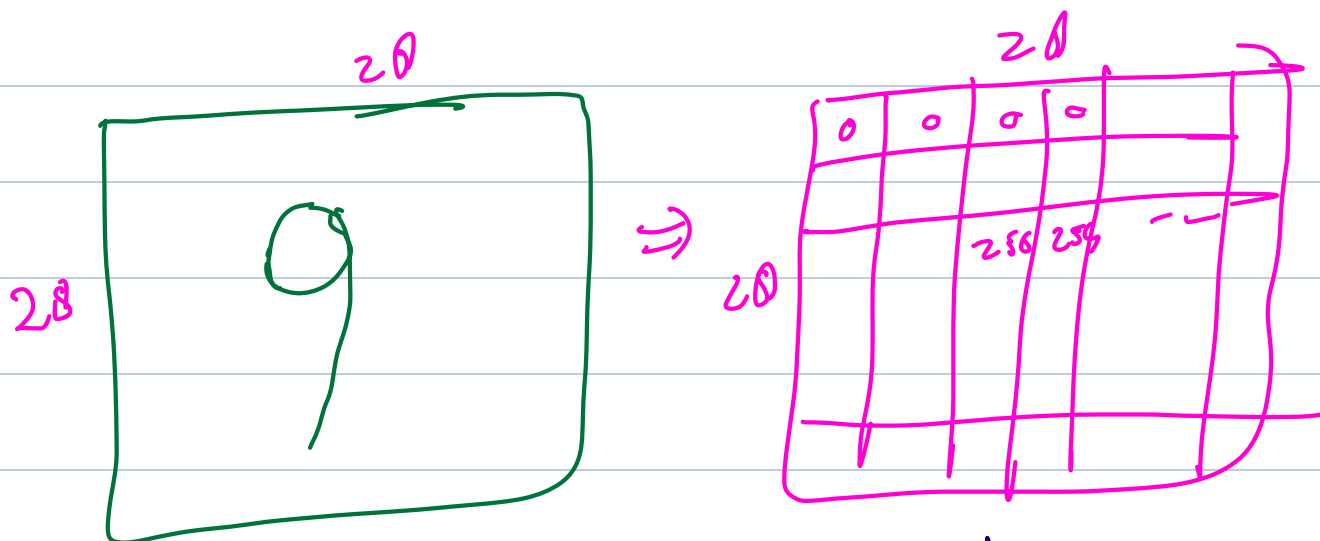
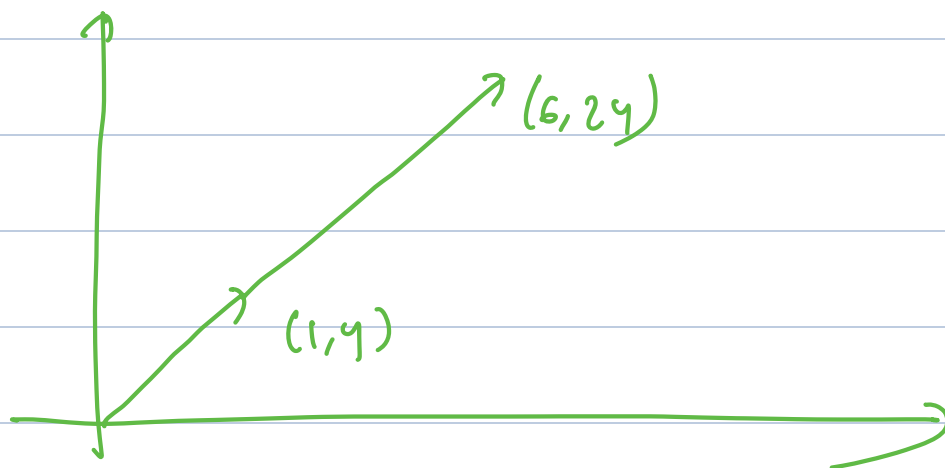


$$V \cdot \vec{u} = \begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 33 \end{bmatrix}$$

$$V \cdot \vec{u} = \begin{bmatrix} -6 & 3 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$


$$= \begin{bmatrix} 6 \\ 24 \end{bmatrix} = 6 \begin{bmatrix} 1 \\ 4 \end{bmatrix}$$



label /
9

$$Ax = \lambda x$$

$$V\vec{v} = \lambda \vec{v}$$

$$Ax - \lambda x = 0$$

$$(A - \lambda)x = 0$$

$\Rightarrow$

$$A = \begin{bmatrix} -5 & 2 \\ -7 & 4 \end{bmatrix}$$

$$\lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\det \left(\begin{bmatrix} -5 & 2 \\ -7 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right) = 0$$

$$\det \left(\begin{bmatrix} -5-\lambda & 2 \\ -7 & 4-\lambda \end{bmatrix} \right) = 0$$

$$-(4-\lambda) \cdot (5+\lambda) + 14 = 0$$

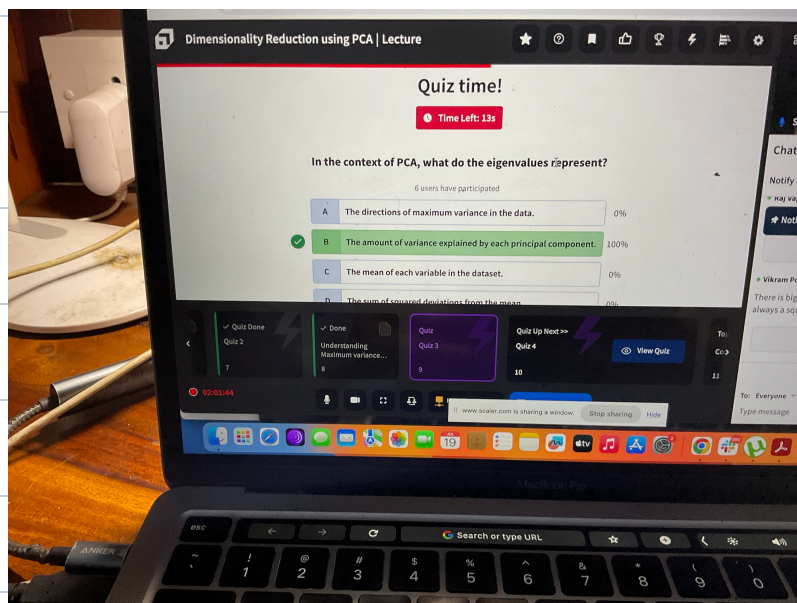
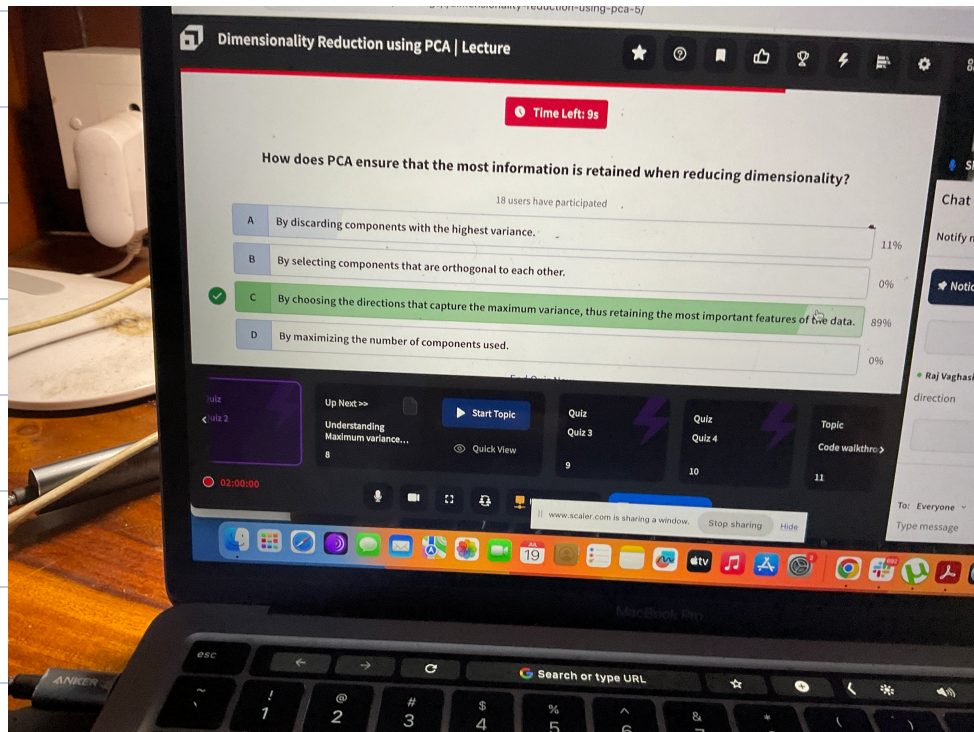
$$\lambda = -3, 2$$

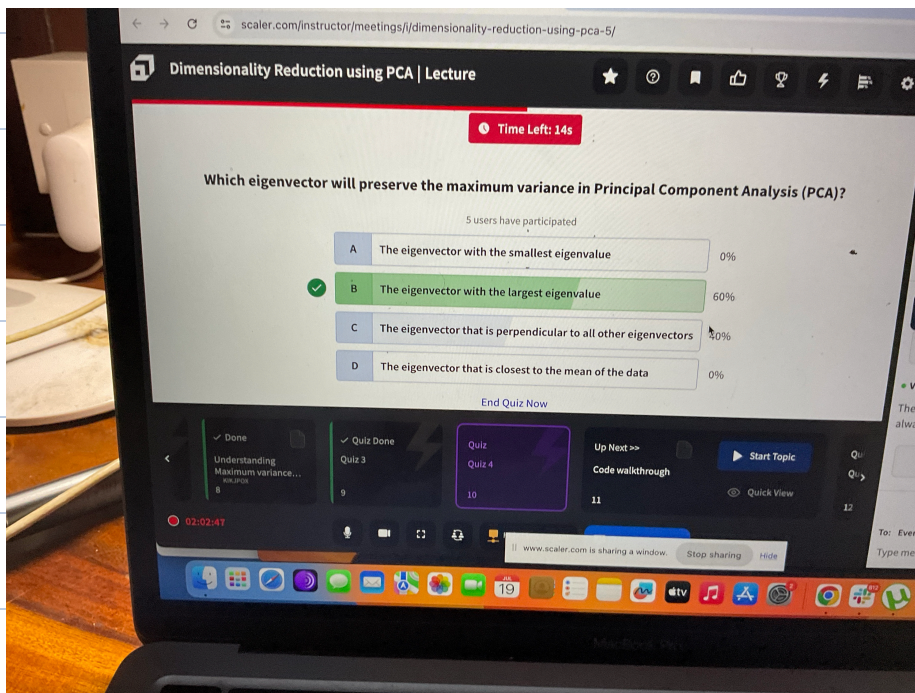
$$Ax = \lambda x$$

$$\lambda x - Ax = 0$$

$$(\lambda I - A) = 0$$

$$2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} -5 & 2 \\ -7 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$





λ_1, λ_2

. explained variance (2)

↓

60%

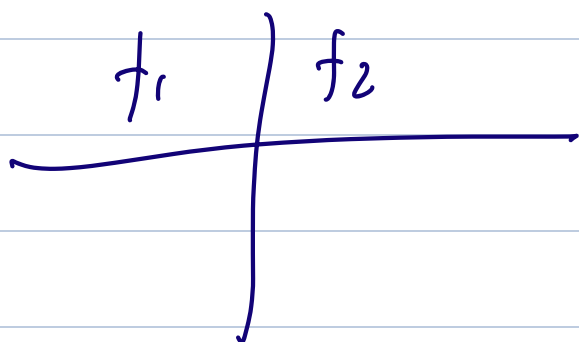
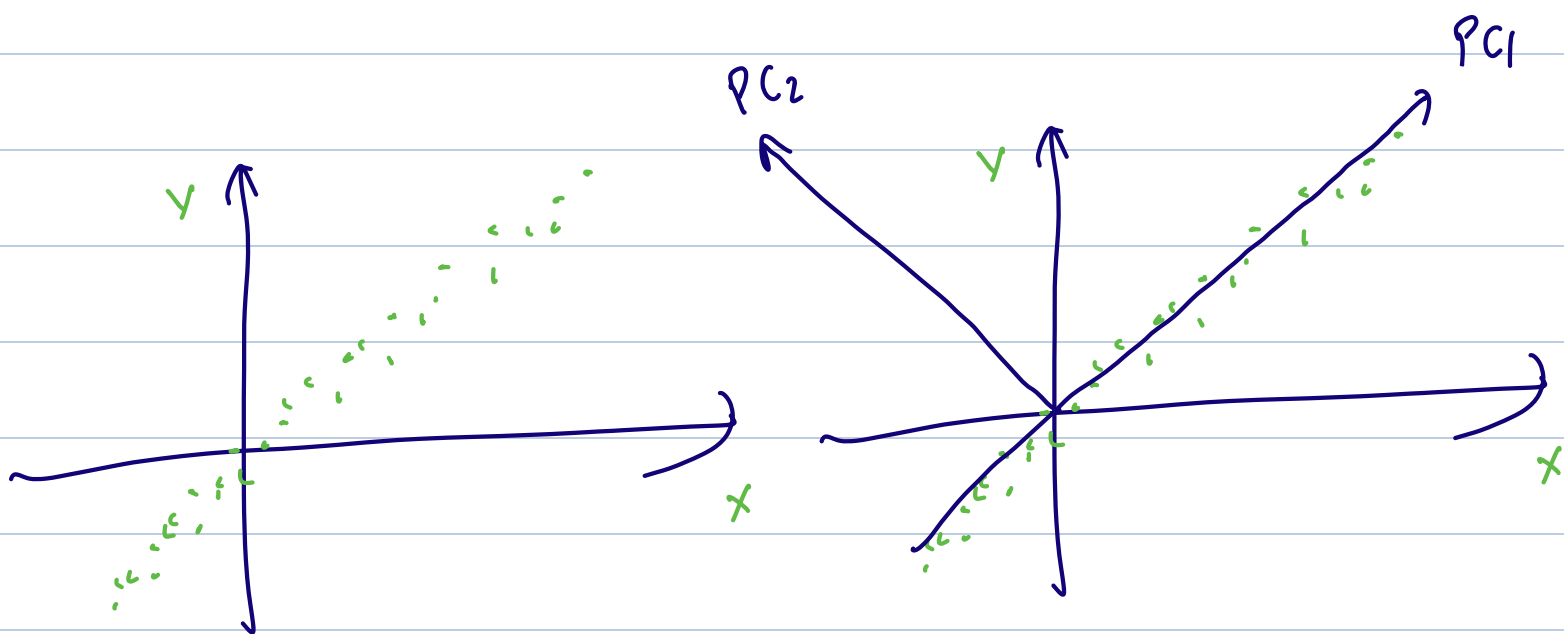
+

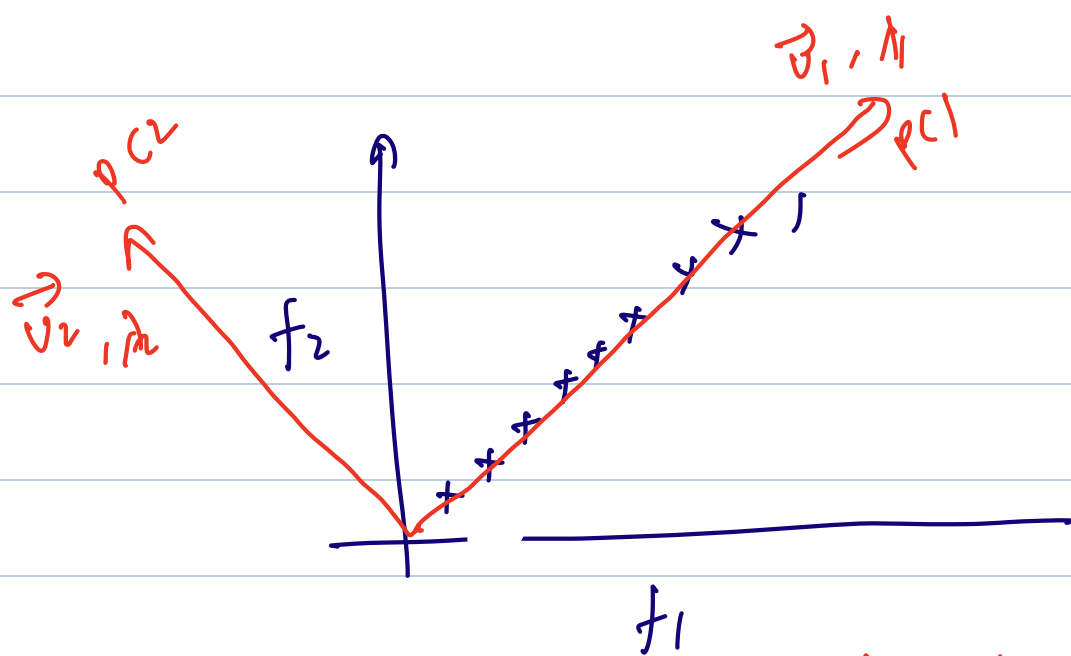
62%

(4)

↓

65%





$$\begin{array}{r} 100 \\ \swarrow \searrow \\ 99 \quad 1 \\ \hline 99 \quad 1 \\ \hline 99+1 \end{array}$$

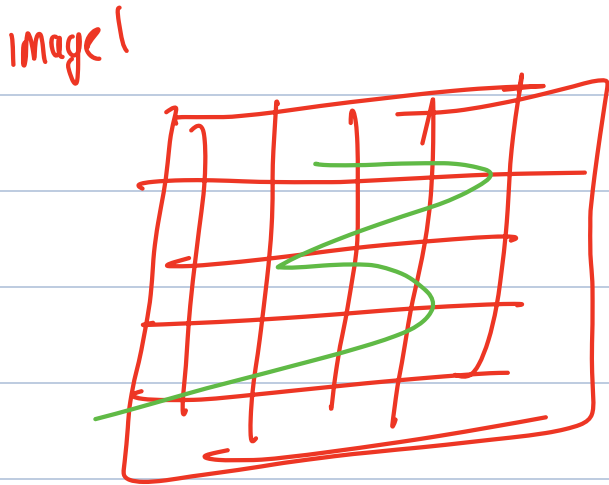
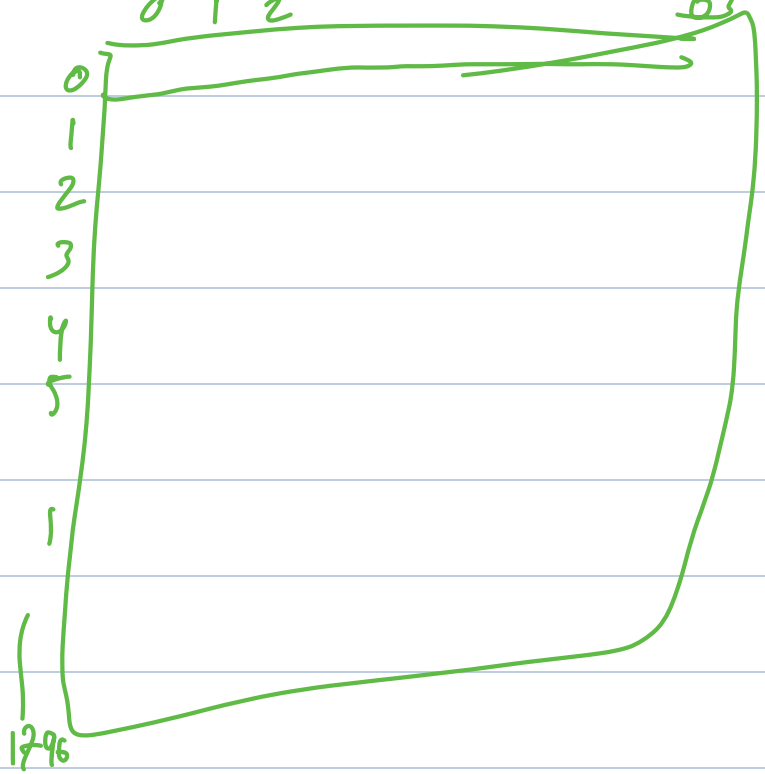
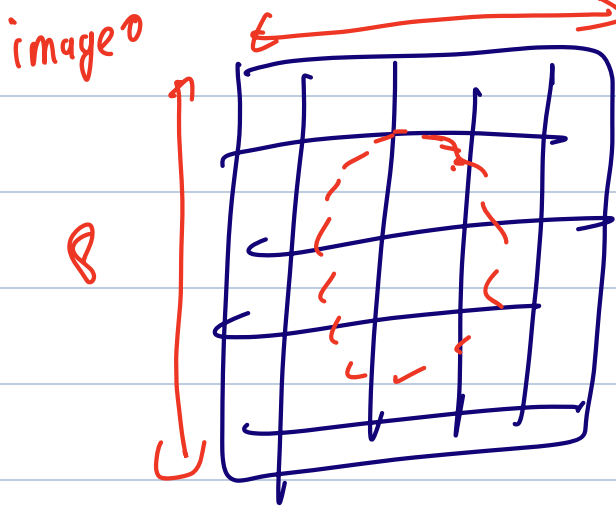
$$\lambda_1, \lambda_2$$

$$\frac{\lambda_1}{\lambda_1 + \lambda_2}, \frac{\lambda_2}{\lambda_1 + \lambda_2}$$

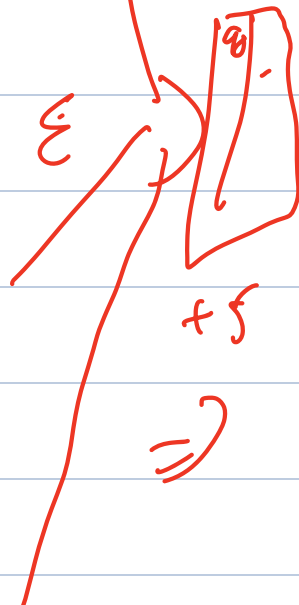
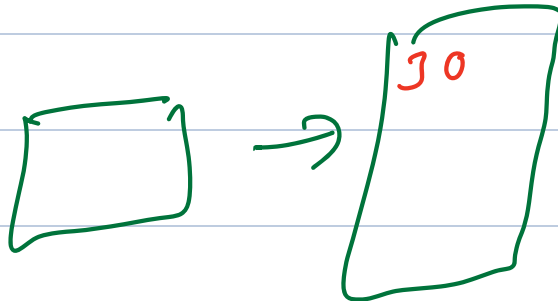
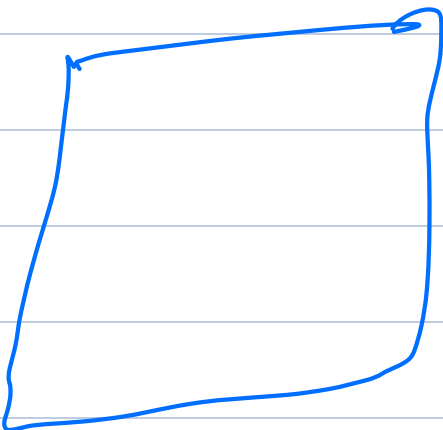
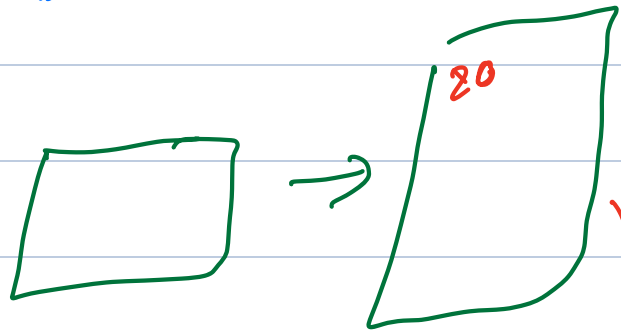
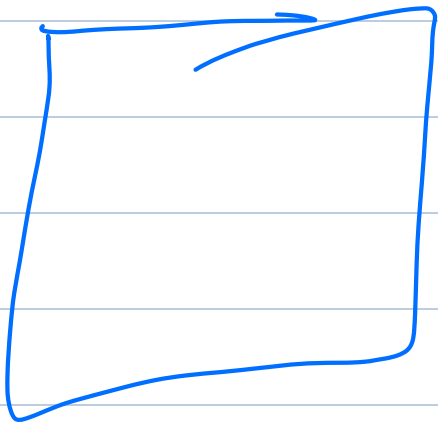
10	→	$10/10+8+6+3+1 \Rightarrow 35\%$
8	→	$\Rightarrow 28\%$
6		$\Rightarrow 21\%$
3		$\Rightarrow 10\%$
1		$\Rightarrow 3\%$

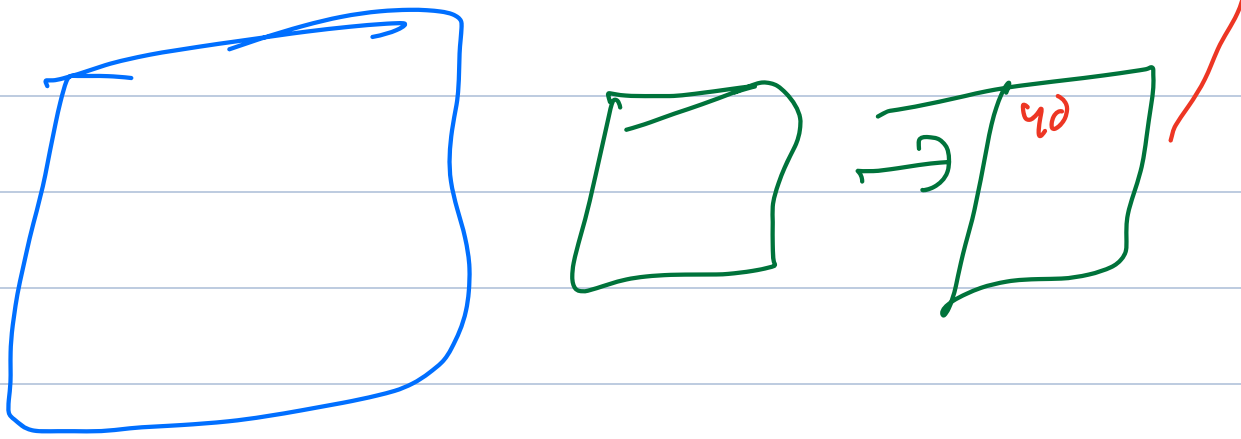
8

17.97



Break : 9:15 PM





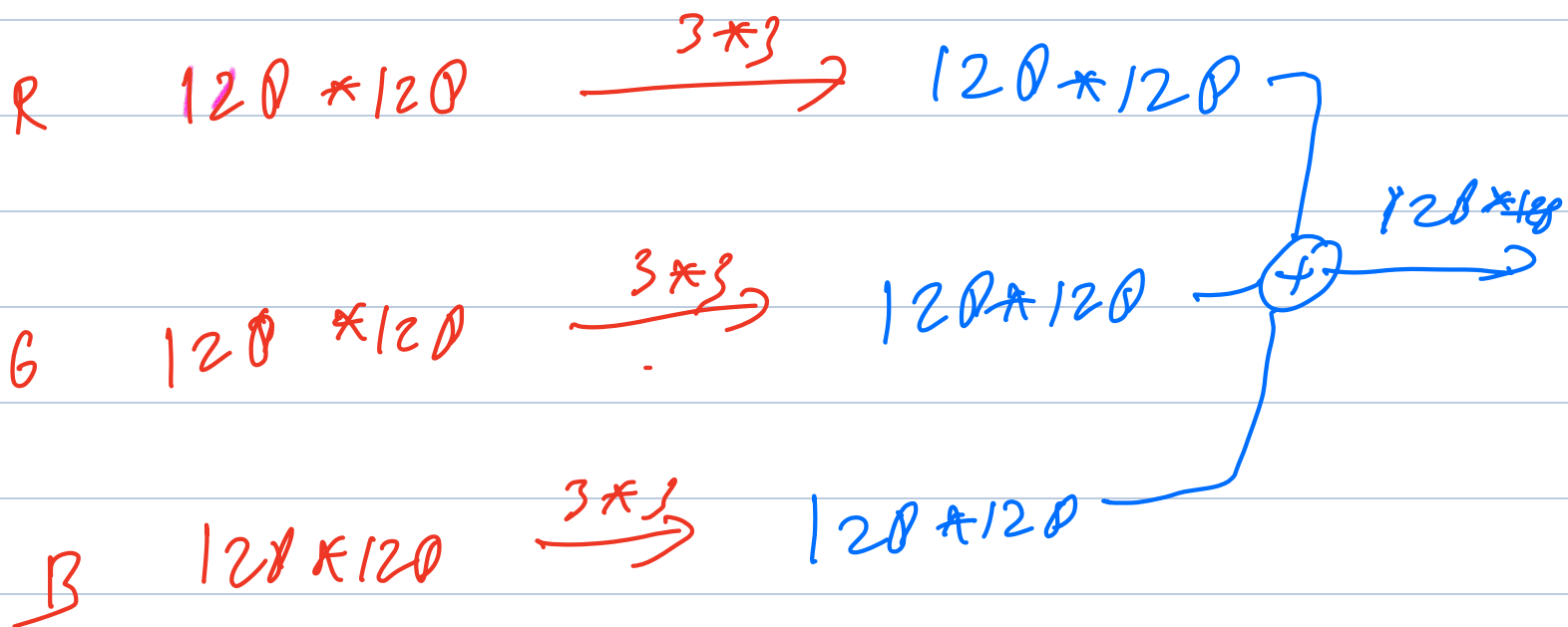
→ Depthwise convolution



Pointwise convolution

$$128 * 128 * 3 \xrightarrow[3 * 3 * 3]{64} 128 * 128 * 64$$

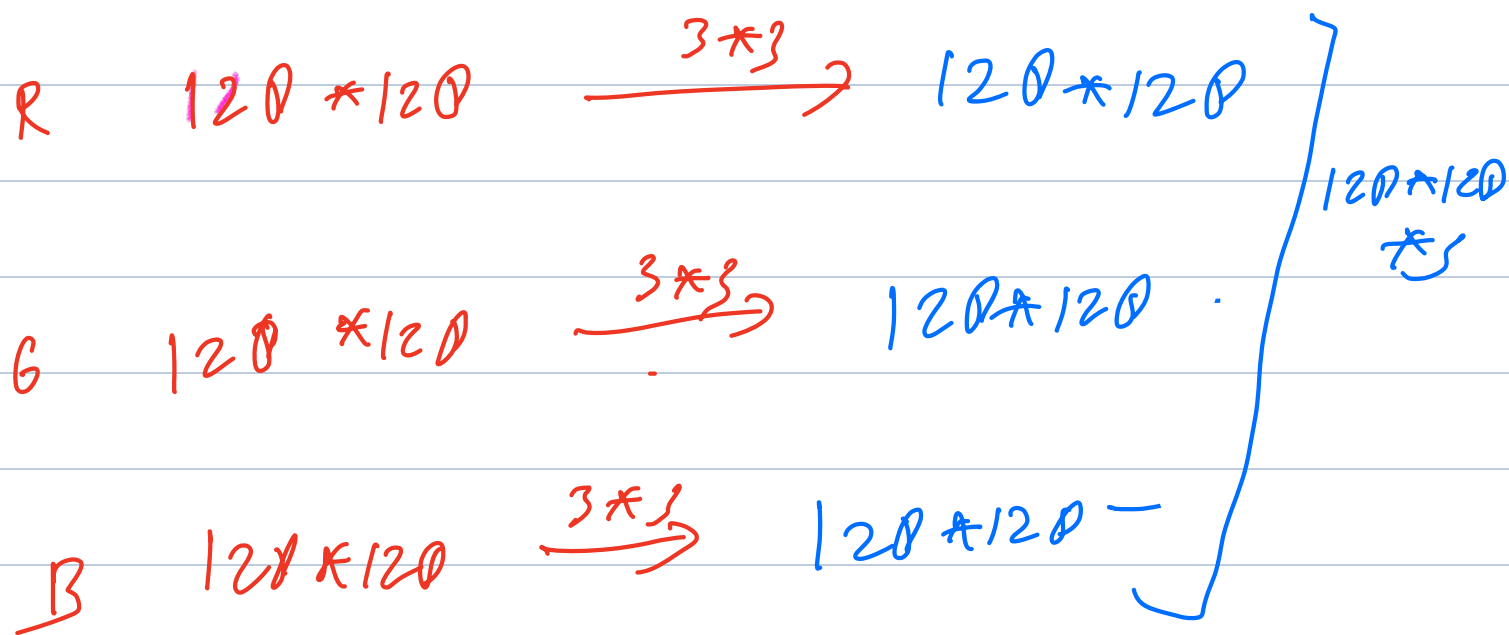
Standard Convolution



Depthwise convolution

Depthwise conv

$$120 * 120 * 3 \xrightarrow[\text{"same"}]{3*3*3} 120 * 120 * 3$$



$$\begin{array}{ccc}
 & DC & \\
 128 * 128 * 3 & \xrightarrow{3 * 3 * 3} & 128 * 128 * 3 \\
 \uparrow & & \uparrow \\
 & \text{-----} &
 \end{array}$$

Pointwise convolution $\rightarrow$

Standard kernel $1 * 1$

standard $\rightarrow 32$

$$128 * 128 * 3 \xrightarrow{3 * 3 * 3} 128 * 128 * 32$$

$$\begin{array}{ccc}
 DC & & \\
 128 * 128 * 3 & \xrightarrow{3 * 3 * 3} & 128 * 128 * 3
 \end{array}$$

$$\begin{array}{c}
 (1 * 1 * 3) \swarrow \begin{array}{l} DC \\ 32 \end{array} \\
 \downarrow
 \end{array}$$

$$128 * 128 * 32$$

Standard

$$K \rightarrow 3 \times 3 \times 512$$

$$\# \text{ Filter} \rightarrow 512$$

total Param $\Rightarrow$

$$(3 \times 3 \times 512 \times 512)$$

$$+ (512)$$

$$\Rightarrow 2.36 \text{ M param}$$

Depthwise conv

step 1] DC

$$K \rightarrow 3 \times 3$$

$$\# \text{ filter} \Rightarrow 512$$

$$\therefore \text{param} \Rightarrow 3 \times 3 \times 512$$

$$\Rightarrow 4608$$

step 2] PC

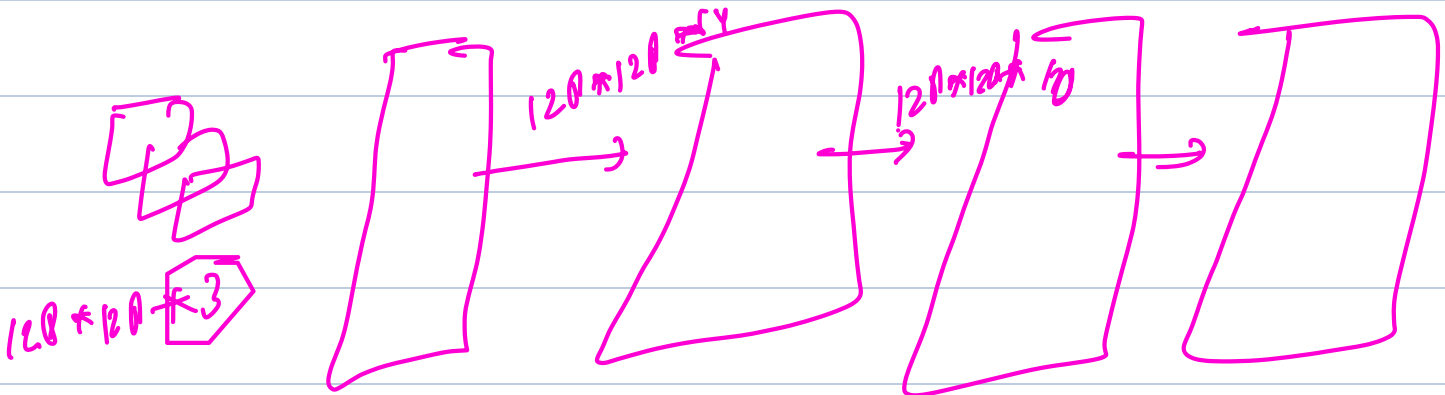
$$K \rightarrow 1 \times 1 \times 512$$

$$\# \text{ Filter} \Rightarrow 512$$

$$\Rightarrow (1 \times 1 \times 512 \times 512) + 512$$

$$\Rightarrow 262656$$

$$\Rightarrow 263 \text{ K}$$



$128 * 128 * 512$
 $\xrightarrow{3 * 3 * 512}$
 512 filter

$128 * 128 * 512$

$128 * 128 * 512 \xrightarrow{3 * 3} 128 * 128 * 512 \xrightarrow{1 * 1 * 512 \text{ # } 512} 128 * 128 * 512$

$512 \left\{ \begin{array}{l} 128 * 128 \xrightarrow{3 * 3} 128 * 128 \\ 128 * 128 \xrightarrow{3 * 3} 128 * 128 \\ \vdots \\ 128 * 128 \xrightarrow{3 * 3} 128 * 128 \end{array} \right.$

